

Name:

Enrolment No:



UPES

End Semester Examination, May 2026

Course: Digital Electronics

Program: B. Tech. CSE

Course Code: ECEG1012

Semester: II
Time: 03 hrs.

Max. Marks: 100

Instructions: Calculator is not allowed

SECTION A
(5Qx4M=20Marks)

S. No.		Marks	CO
Q 1	Design an XNOR gate using NOR gate.	4	CO1
Q 2	Illustrate working of a full subtractor using truth table and circuit diagram.	4	CO2
Q 3	Explain demultiplexing of address data bus in 8085 with diagram.	4	CO4
Q 4	Design an S-R latch using NOR gates and explain working with truth table.	4	CO2
Q 5	Define the following parameters of Digital to Analog Converters (DACs). (a) Resolution (b) Accuracy (c) Settling Time (d) Offset Voltage	4	CO3

SECTION B
(4Qx10M= 40 Marks)

Q 6	A combinational circuit receives two 2-bit binary numbers $A = A_1A_0$ and $B = B_1B_0$ and compares them to determine whether $A > B$, $A = B$, or $A < B$. (a) Construct the truth table showing all possible combinations of inputs A_1, A_0, B_1, B_0 and the corresponding outputs $A > B$, $A = B$, and $A < B$. (3 marks) (b) Derive the Boolean expressions for the outputs $A > B$, $A = B$, and $A < B$ using Boolean algebra or Karnaugh maps. (5 marks) (c) Design the corresponding logic circuit for the 2-bit comparator using basic logic gates. (2 marks)	10	CO2
Q 7	Design a 3-bit Synchronous Down Counter using JK flip-flops. (a) Draw the state diagram for the 3-bit synchronous down counter. (2 marks) (b) Determine the excitation inputs for each JK flip-flop using the excitation table. (2 marks) (c) Derive the Boolean expressions for the inputs of each flip-flop. (4 marks)	10	CO2

	(d) Draw the logic circuit diagram of the 3-bit synchronous down counter. (2 marks)		
Q 8	Explain the following five addressing modes of the 8085 microprocessor with suitable instruction examples. Also explain how the operand is accessed in each mode. (a) Immediate Addressing Mode (b) Register Addressing Mode (c) Direct Addressing Mode (d) Register Indirect Addressing Mode (e) Implied Addressing Mode	10	CO4
Q 9	(a) Explain working of R-2R ladder type DAC with suitable circuit diagram. (6 marks) (b) An 6-bit DAC has a step size of 50 mV. Determine the full scale output voltage and percentage resolution. (4 marks) OR Explain working of the Dual-Slope type ADC with suitable diagram.	10	CO3

SECTION-C
(2Qx20M=40 Marks)

Q 10	(a) Convert SR Flip Flop to T Flip Flop. (5 marks) (b) Obtain the minimal expression using tabular method: (15 marks) $f = \sum_m (1, 2, 3, 5, 6, 7, 8, 9, 12, 13, 15)$	20	CO2
Q 11	(a) Design and illustrate working of a 4-bit Ring counter using state diagram, sequence table and logic diagram. (5 marks) (b) Design a Mod-10 Asynchronous counter using T Flip Flop. (15 marks) OR (a) Explain working of 4 bit serial-in, parallel-out shift register with suitable diagram (5 marks). (b) Design a sequence generator to generate the sequence of odd numbers: $1 \rightarrow 3 \rightarrow 5 \rightarrow 7 \rightarrow$ repeat continuously. (i) Construct state diagram the given sequence. (2 marks) (ii) Using D flip-flops, determine excitation table for each flip-flop. (3 marks) (iii) Derive Boolean expressions for the flip-flop inputs using K-map simplification. (5 marks) (iv) Draw the logic circuit diagram of the sequence generator and briefly explain its operation. (5 marks)	20	CO2